

1.15 Use both the equations below for all waves:

wave speed (metre/second, m/s) = frequency (hertz, Hz) × wavelength (metre, m)

$$v = f \times \lambda$$

wave speed (metre/second, m/s) = distance (metre, m) / time (second, s)

$$v = \frac{x}{t}$$

Topic 2

The electromagnetic spectrum

- 2.1 Demonstrate an understanding of how Herschel and Ritter contributed to the discovery of waves outside the limits of the visible spectrum
- 2.2 Demonstrate an understanding that all electromagnetic waves are transverse and that they travel at the same speed in a vacuum
- 2.3 Describe the continuous electromagnetic spectrum including (in order) radio waves, microwaves, infrared, visible (including the colours of the visible spectrum), ultraviolet, X-rays and gamma rays
- 2.4 Demonstrate an understanding that the electromagnetic spectrum is continuous from radio waves to gamma rays, but the radiations within it can be grouped in order of decreasing wavelength and increasing frequency
- 2.5 Demonstrate an understanding that the potential danger associated with an electromagnetic wave increases with increasing frequency
- 2.6 Relate the harmful effects, to life, of excessive exposure to the frequency of the electromagnetic radiation, including:
 - a microwaves: internal heating of body cells
 - b infrared: skin burns
 - c ultraviolet: damage to surface cells and eyes, leading to skin cancer and eye conditions
 - d X-rays and gamma rays: mutation or damage to cells in the body

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- 2.7 Describe some uses of electromagnetic radiation
 - a radio waves: including broadcasting, communications and satellite transmissions
 - b microwaves: including cooking, communications and satellite transmissions
 - c infrared: including cooking, thermal imaging, short range communications, optical fibres, television remote controls and security systems
 - d visible light: including vision, photography and illumination
 - e ultraviolet: including security marking, fluorescent lamps, detecting forged bank notes and disinfecting water
 - f X-rays: including observing the internal structure of objects, airport security scanners and medical X-rays
 - g gamma rays: including sterilising food and medical equipment, and the detection of cancer and its treatment
- 2.8 Recall that ionising radiations are emitted all the time by radioactive sources
- 2.9 Describe that ionising radiation includes alpha and beta particles and gamma rays and that they transfer energy

Topic 3

Waves and the Universe

- 3.1 Recall that the Solar System is part of the Milky Way galaxy
- 3.2 Describe a galaxy as a collection of stars
- 3.3 Recall that the Universe includes all of the galaxies
- 3.4 Compare the relative sizes of and the distances between the Earth, the Moon, the planets, the Sun, galaxies and the Universe
- 3.5 Describe the use of other regions of the electromagnetic spectrum by some modern telescopes
- 3.6 Describe the methods used to gather evidence for life beyond Earth, including space probes, soil experiments by landers, Search for Extraterrestrial Intelligence (SETI)
- 3.7 Demonstrate an understanding of the impact of data gathered by modern telescopes on our understanding of the Universe, including:
 - a the observation of galaxies because of improved magnification
 - b the discovery of objects not detectable using visible light
 - c the ability to collect more data
- 3.8 *Construct a simple spectrometer, from a CD or DVD, and use it to analyse common light sources*
- 3.9 Explain why some telescopes are located outside the Earth's atmosphere

- 3.10 **Analyse data provided to support the location of telescopes outside the Earth's atmosphere**
- 3.11 Describe the evolution of stars of similar mass to the Sun through the following stages:
 - a nebula
 - b star (main sequence)
 - c red giant
 - d white dwarf
- 3.12 Describe the role of gravity in the life cycle of stars
- 3.13 **Describe how the evolution of stars with a mass larger than the Sun is different, and may end in a black hole or neutron star**
- 3.14 Demonstrate an understanding of the Steady State and Big Bang theories
- 3.15 Describe evidence supporting the Big Bang theory, limited to red-shift and the cosmic microwave background (CMB) radiation
- 3.16 Recognise that as there is more evidence supporting the Big Bang theory than the Steady State theory, it is the currently accepted model for the origin of the Universe
- 3.17 Describe that if a wave source is moving relative to an observer there will be a change in the observed frequency and wavelength
- 3.18 **Demonstrate an understanding that if a wave source is moving relative to an observer there will be a change in the observed frequency and wavelength**
- 3.19 **Describe the red-shift in light received from galaxies at different distances away from the Earth**
- 3.20 **Explain why the red-shift of galaxies provides evidence for the Universe expanding**
- 3.21 **Explain how both the Big Bang and Steady State theories of the origin of the Universe both account for red-shift of galaxies**
- 3.22 **Explain how the discovery of the CMB radiation led to the Big Bang theory becoming the currently accepted model**

Topic 4

Waves and the Earth

- 4.1 Recall that sound with frequencies greater than 20 000 hertz, Hz, is known as ultrasound
- 4.2 Describe uses of ultrasound, including:
 - a sonar
 - b communication between animals
 - c foetal scanning
- 4.3 Calculate depth or distance from time and velocity of ultrasound

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- 4.4 Recall that sound with frequencies less than 20 hertz, Hz, is known as infrasound
- 4.5 Describe uses of infrasound, including:
 - a communication between animals
 - b detection of animal movement in remote locations
 - c detection of volcanic eruptions and meteors
- 4.6 Recall that seismic waves are generated by earthquakes or explosions
- 4.7 *Investigate the unpredictability of earthquakes, through sliding blocks and weights*
- 4.8 Explain why scientists find it difficult to predict earthquakes and tsunami waves even with available data
- 4.9 Recall that seismic waves can be longitudinal (P) waves and transverse (S) waves and that they can be reflected and refracted at boundaries between the crust, mantle and core